

Week 4 Project Summary

Week 4 marked the start of the **HealthConnect Clinic Experience Lab**. The Data Science task was to define a machine learning problem that could help the clinic reduce missed appointments. The main goal was to assess whether the available appointment data could support a no-show prediction model.

The dataset contains 5,000 appointment records with information about patient demographics, appointment details, booking history, previous no-shows, reminders, distance to the clinic, estimated waiting time, and appointment outcomes.

The initial assessment showed that the dataset is generally suitable for modelling. No duplicate records or major consistency problems were identified. Some missing values were present, but most were limited. Missing values in `reminder_channel` were found to have a logical reason because the channel is not recorded when no reminder is sent.

The proposed machine learning problem is binary classification. The model will predict whether an appointment will be Attended or No-Show. Cancelled appointments will initially be excluded because cancellation represents a different outcome from failing to attend.

Potential predictors include age, gender, appointment type, appointment day and time, booking lead days, previous appointments, previous no-shows, reminder information, distance to the clinic, and estimated waiting time.

Important considerations include preventing data leakage, handling repeated patients correctly, treating missing values appropriately, and evaluating possible bias in demographic variables.

The next phase will focus on deeper analysis, preprocessing, feature engineering, and the development of an initial no-show prediction model. This follows the Week 4 requirement to establish a clear foundation for the work that continues in Week 5.

Excited to see what's coming

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